

Crystal Chang

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EDUCATION

National Taiwan University

[2022-2027]

Double Major: Information Management / Political Science

GPA: 4.16/4.3

Award: Presidential Award, Dean's List Award

WORK EXPERIENCE

RD Intern | Microsoft

[Sep 2025-Now]

- Built a Spark-based distributed geospatial data processing pipeline for GeoJSON-to-PMTiles transformation, including environment setup, pipeline integration, and large-scale performance testing
- Optimized distributed parallel-processing workflows, reducing geospatial data transformation time by 20%
- Developed a production-level WFS (Web Feature Service) frontend system using React and TypeScript, supporting feature querying, spatial filtering, and interactive map visualization workflows

AI Engineer Intern | Trend Micro

[Jul, 2025-Aug, 2025]

- Developed an AI-driven proactive monitoring system that detects service alerts from logs and provides recommended solutions, reducing average resolution time from 3 hours to 10 minutes for the SRE team
- Implemented an MCP server framework using FastMCP, including the tools from data preprocessing with Dify and N8N, RAG for finding corresponding documents, to LLM-based summary analysis
- Built a Streamlit-based chatbot interface that connects with the MCP server via an agent framework

Research Assistant | EVA Airways Corp & NTU IM Lab

[Feb, 2025-Dec, 2025]

- Collaborated with NTU IM Lab to develop an AI-driven pilot performance analysis system, coordinating with lab researchers and EVA Airways specialists to align technical requirements and progress
- Developed a React-based frontend, implementing data visualizations of machine learning model outputs and building UI components from wireframes

Business Analysis Intern | Jubo (Smart Ageing Tech)

[Jun, 2024-Aug, 2024]

- Developed a Python-based financial automation system to streamline debt write-offs and track customer accounts receivable across diverse SaaS subscription plans, increasing efficiency in financial tracking
- Built an API for the automation system by integrating Python with cloud storage, facilitating direct file uploads for users and enabling the finance team to efficiently monitor and maintain accurate financial data
- Designed and implemented a Power BI dashboard for real-time financial insights, utilizing Python to automate data updates, supporting the sales team to identify customer trends and make data-driven decisions

PROJECT EXPERIENCE

Health Management System | Database Management & Full Stack Design

[Sep, 2024-Feb, 2025]

- Developed a full-stack health management system for dietary tracking and nutrient analysis with role-based access using Next.js
- Designed a PostgreSQL database with 100K+ records, integrating official government nutrient datasets and implementing triggers for scalability and maintainability

Machine Learning for Predicting Fubon Insurance Cases

[Dec, 2023-Jun, 2024]

- Built a predictive model with Scikit-learn (logistic regression, polynomial features, regularization, cross-validation) in collaboration with Fubon Life Insurance
- Achieved 84% accuracy, improving insurance case categorization and allocation efficiency for solicitors

SKILLS

Programming & Systems: C++, Python, SQL, Web Applications, AI/ML Applications, Data Analysis, Database

Tools & Platforms: Azure, Power BI, Dify, N8N, React, Next.js, Google Workspace (Docs, Slides, Sheets)

Languages: English, Mandarin, Shanghai Dialect